

A shared release produces exclusive fringe-weighted events

An autonomous classical jump receiver can produce exactly one eventual event with probabilities equal to normalized interference energies. It does so by admitting one shared release and prescribing rates linear in those energies. Finite observation time reveals a signal-dependent efficiency; spatially separated local releases require a communication or common-control mechanism. Neither construction selects a universal action constant.

Status: Q11 exploratory model and written derivation. This is an effective stochastic receiver, not a microscopic Hamiltonian derivation. Its discrete release states, stored energy and rate law are explicit model inputs.

1. Autonomous competition and energy accounting

First load the two classical work records from Q08 into ideal storage:

$$e_+ + e_- = E > 0, \quad e_{\pm} = E(1 \pm z)/2, \quad -1 \leq z \leq 1.$$

The following receiver stage has constant stored inputs. Let R be a ready state and C_+, C_- be two absorbing record states. Choose a transduction constant $\alpha > 0$ of units (energy time)⁻¹ and transitions

$$R \xrightarrow{r_+ = \alpha e_+} C_+, \quad R \xrightarrow{r_- = \alpha e_-} C_-. \quad (1)$$

This time-homogeneous Markov model is autonomous after preparation. Its absorbing architecture stipulates that the first channel consumes the single permission to release; no second transition remains. The stored signal energies act as kinetic controls and are not automatically the emitted record energy.

For explicit energy bookkeeping, put one releasable amount $\Delta > 0$ in a central reservoir. Either transition lowers that reservoir by Δ and increases the selected output's energy plus heat by Δ . Signal storage is unchanged in this idealization. Resetting the released reservoir requires replacement energy Δ , or retrieval from an explicitly included output store. The architecture supplies single use; energy conservation alone does not imply the state graph (1). Permanent storage, reset efficiency and a physical implementation of the rate control remain additional premises.

2. Exact probabilities and their limit

Write $\Gamma = r_+ + r_- = \alpha E$. The survival equation is $\dot{p}_R = -\Gamma p_R$, $p_R(0) = 1$. Integrating each exit flux gives

$$p_R(t) = e^{-\Gamma t}, \quad p_{\pm}(t) = \frac{r_{\pm}}{\Gamma} (1 - e^{-\Gamma t}). \quad (2)$$

Double records have probability zero. Conditional on one event by any fixed $t > 0$, or unconditionally after waiting indefinitely,

$$\Pr(C_{\pm} \mid \text{event}) = \frac{e_{\pm}}{E} = \frac{1 \pm z}{2}. \quad (3)$$

These fringe weights follow from the assumed linear hazards and competition; no quantum probability rule was used to obtain them. They also do not establish quantum state structure. Replacing the hazards by $\alpha_p e_{\pm}^p$, with the corresponding units and $p > 0$, gives weights $e_{\pm}^p / (e_+^p + e_-^p)$. Exclusivity therefore does not select linearity.

If the incident displacement is attenuated by $\eta > 0$, then $e_{\pm}$ scale by η^2 and z is unchanged. Equation (3) persists, but at a fixed deadline T the efficiency becomes $1 - \exp(-\alpha \eta^2 E T)$ and tends to zero. Requiring efficiency at least $1 - \varepsilon$, $0 < \varepsilon < 1$, gives

$$E \geq \frac{\log(1/\varepsilon)}{\alpha T}. \quad (4)$$

At fixed carrier frequency ω in Q08's narrow-band regime, the associated wave-action bound is approximately $\log(1/\varepsilon)/(\alpha T \omega)$. It retains receiver coupling, deadline, frequency and error tolerance. The indefinite-wait and zero-signal limits do not commute. At $E=0$ there is no event.

3. What spatial separation changes

Equation (1) is consistent as a co-located central arbitration model: both stored inputs are available before the release. Sending the output to a remote record then takes finite travel time. It is not an instantaneous inhibition law for distant detectors that already register events independently.

To test that alternative, let the two sites initially have independent Poisson trigger clocks with rates r_+, r_- . After the first event, an inhibition signal reaches the other site after a fixed delay $\delta > 0$. Each site can fire only once. On an observation interval long enough to include that delay, memorylessness gives the exact eventual double-event probability

$$P_2 = \frac{r_+}{\Gamma}(1 - e^{-r_-\delta}) + \frac{r_-}{\Gamma}(1 - e^{-r_+\delta}). \quad (5)$$

It is positive whenever both rates and the delay are positive. If their separation is d and inhibition speed is at most c , $\delta \geq d/c$ in this specified architecture. In the balanced case $r_+ = r_- = \Gamma/2$,

$$P_2 = 1 - e^{-\Gamma\delta/2}. \quad (6)$$

Combining a first-event deadline T with $P_2 \leq \varepsilon_2$ and $\Pr(\text{no first event by } T) \leq \varepsilon_0$ requires

$$\frac{\log(1/\varepsilon_0)}{T} \leq \Gamma \leq \frac{-2\log(1 - \varepsilon_2)}{\delta}. \quad (7)$$

The double-event criterion here counts events during the full inhibition window, even when it extends past T . Short-gate censoring would change that criterion. Equation (7) is a timing compatibility condition, not an action gap or a bound for every classical architecture. A central arbiter, prior correlated readiness, or a different propagation law changes a premise.

4. Decision

Shared-resource competition successfully turns classical fringe energies into exclusive conditional event probabilities in a declared model. The resource preparation and linear kinetic response do substantive work. Replacing central arbitration by local firing with delayed inhibition adds a calculable efficiency/coincidence trade-off. Both mechanisms retain adjustable energy and clock scales.

After Q09–Q11, park further detector tuning. The decisive next construction is a pair of independently prepared local receivers with no communication during the gate: ask whether independent monotone local response to a common classical pulse energy permits suppressed coincidences without selection of gates. This changes the common-resource premise rather than refining another rate constant. Keep source conditioning and detector independence explicit; any resulting exclusion must be limited to that stated class.

Source boundary

D. T. Gillespie, *Exact stochastic simulation of coupled chemical reactions*, Journal of Physical Chemistry **81**, 2340–2361 (1977), DOI, is the primary discovery lead for the standard competing-hazard construction. Coverage here is one query and publisher metadata, not a full-text audit

or simulation. Equations (2)–(7) are obtained directly by integrating exponential survival probabilities. Independent written review and bounded librarian comparison are required before accepted-ledger promotion. No numerical or symbolic scripts were used.